

CUBICLE NO 8

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Day 1: Arduino Uno Board Analysis

- **Date:** 17-07-2025
- **Domain:** Embedded Systems and IoT
- **Topic:** Analysis of Arduino Uno Board

Objective

To understand the architecture, features, and limitations of the Arduino Uno board by analyzing its components, pin configuration, functionalities, datasheet, and real-world applications.

What is Arduino Uno?

Arduino Uno is an open-source microcontroller board based on the ATmega328P microcontroller. It is one of the most popular development boards used in electronics, automation, robotics, and embedded system applications due to its simplicity and wide community support.

Key Specifications

- **Microcontroller:** ATmega328P
- **Operating Voltage:** 5V
- **Input Voltage (recommended):** 7–12V
- **Digital I/O Pins:** 14 (of which 6 can provide PWM output)
- **Analog Input Pins:** 6
- **Flash Memory:** 32 KB (ATmega328P)
- **SRAM:** 2 KB
- **EEPROM:** 1 KB
- **Clock Speed:** 16 MHz

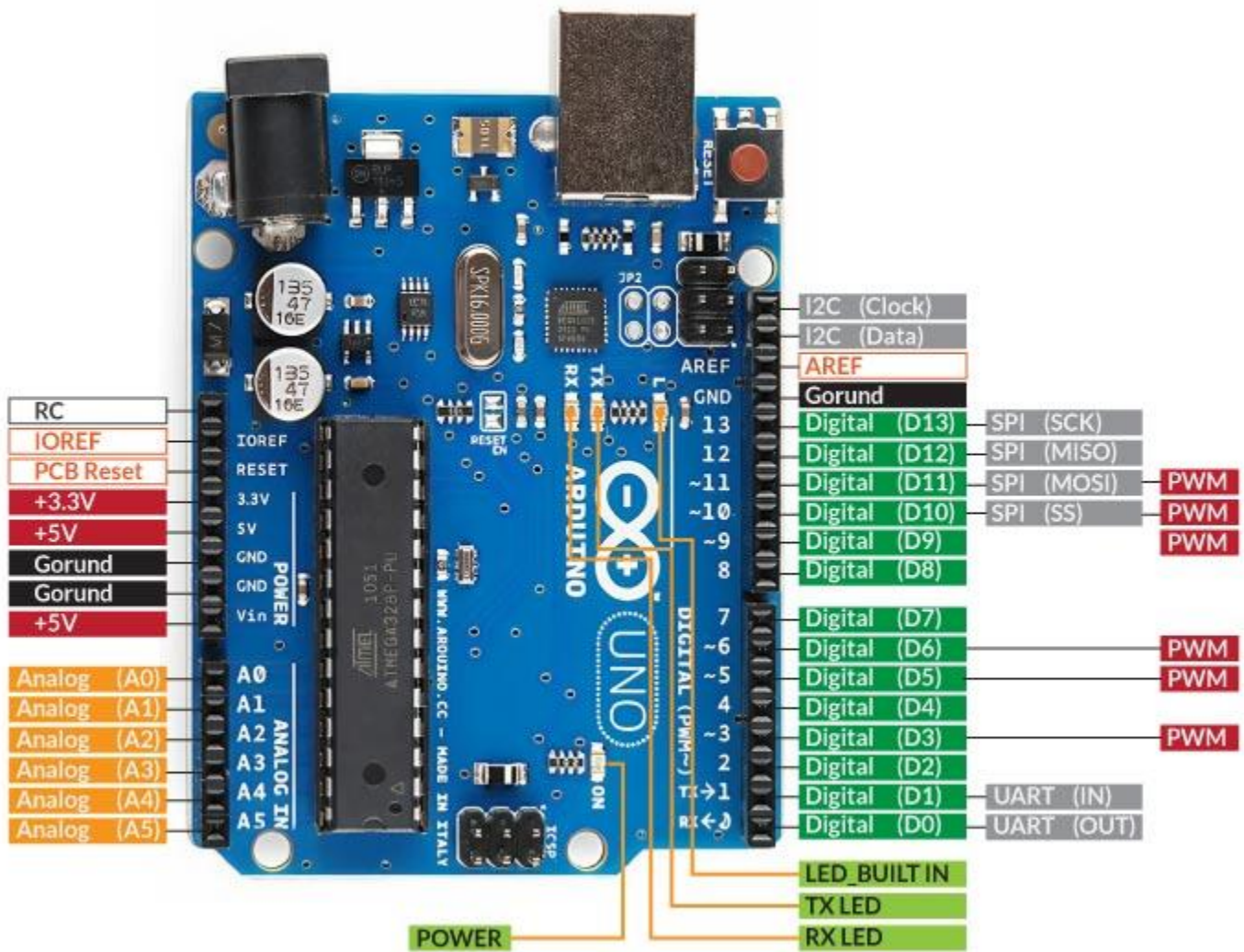
- **USB Interface:** Type-B USB

Pin Configuration

Pin Type	Quantity	Description
Digital I/O	14	General-purpose digital input/output pins
PWM Pins	6	Used for analog output via analogWrite()
Analog Input	6	For reading analog sensor values (A0 to A5)
Power Pins	6	Includes 5V, 3.3V, GND, Vin
Reset Pin	1	Used to reset the microcontroller manually
TX/RX Pins	2	Serial communication indicators (Pin 0 & 1)

Block Diagram & Board Overview

- **ATmega328P MCU** at the core for computation and control
- **USB Interface** for programming and power
- **Voltage Regulator** to manage external input voltage
- **Crystal Oscillator** (16 MHz) for timing operations
- **ICSP Header** for bootloader burning or programming
- **LED on Pin 13** for testing and diagnostics



Applications

- Prototyping IoT devices
- Home automation systems
- Educational and learning kits
- Sensor and actuator interfacing
- Robotics and embedded control

Advantages

- Open-source and beginner-friendly
- Easy-to-use Arduino IDE for programming
- Extensive online support and community
- Available libraries for a wide range of components
- Affordable and readily available

Disadvantages

- Limited processing power (8-bit)
- Low memory (32 KB flash)
- No built-in Wi-Fi or Bluetooth (external modules required)
- Slower clock speed compared to modern microcontrollers
- Not suitable for complex/real-time applications

Limitations

- Cannot multitask or run real-time OS
- Limited GPIO for high-end projects
- Not ideal for image/video processing tasks

- USB communication is relatively slow for high-speed data transfer

Datasheet Analysis Summary

ATmega328P Datasheet Highlights:

- Supports 1.8–5.5V operation
- 23 GPIO lines
- 10-bit ADC with 6 input channels
- USART, SPI, I2C communication protocols supported
- Watchdog timer and power-saving modes
- 3 timers (1x16-bit, 2x8-bit)

Link : <https://docs.arduino.cc/resources/datasheets/A000066-datasheet.pdf>

Arduino IDE Download and Installation

Steps Followed:

1. **Visited** the official Arduino website:
<https://www.arduino.cc/en/software>
2. **Downloaded** the Arduino IDE (Windows Installer version for Windows 10/11).
3. **Installed** the Arduino IDE using the installer:
 - Accepted license agreement
 - Selected all installation options (drivers, shortcuts, etc.)
 - Chose installation path
 - Completed setup
4. **Launched** Arduino IDE successfully.
5. **Verified** installation by navigating to:
Tools > Board > Arduino Uno and ensured the correct COM port was detected.

Learning Outcomes

- Understood the complete structure and capabilities of the Arduino Uno board.
- Explored the role of ATmega328P microcontroller and pin functions.
- Identified advantages and drawbacks of using Arduino Uno.
- Installed and configured Arduino IDE successfully for future coding sessions.